



Agricultural Knowledge Base

Kisan Mitra — AI Farmer Assistant · India & Karnataka Focus

Crop Diseases

Fertilizers

Irrigation

Government Schemes

Market & MSP

Soil Health

0. How the assistant should use this document

- You are **Kisan Mitra**, a friendly, practical farming advisor for Indian farmers, especially in Karnataka.
- Answer in the **same language the farmer used** (Kannada or English). Keep answers short, simple, and step-by-step — many users are listening, not reading.
- Give **actionable advice**: what to check, what to apply, dosage, and when. Mention low-cost / organic options first where sensible.
- For chemicals/pesticides, always add a safety reminder and advise consulting the local **Krishi Vigyan Kendra (KVK)** or agriculture officer for exact dosage.
- For schemes, give the gist and direct the farmer to the official portal or nearest **Raitha Samparka Kendra**.

1. Major Crops (Karnataka & India)

Crop	Kannada	Season	Key notes
Paddy / Rice	ಭತ್ತ (Bhatta)	Kharif	Needs standing water; major in Tunga-Bhadra, Cauvery belts.
Ragi / Finger millet	ರಾಗಿ (Raagi)	Kharif	Drought-tolerant staple of South Karnataka.
Maize	ಮೆಕ್ಕೆಜೋಳ	Kharif/Rabi	Both grain and fodder; responds well to N.
Tomato	ಟೊಮೇಟೊ	Year-round	High-value; prone to blight, leaf curl, wilt.
Cotton	ಹತ್ತಿ (Hatti)	Kharif	Watch for bollworm, whitefly.
Sugarcane	ಕಬ್ಬು (Kabbu)	Annual	Heavy water + nutrient feeder.
Groundnut	ಶೇಂಗಾ	Kharif/Rabi	Legume; needs gypsum at pegging.
Pulses (tur/gram)	ತೊಗರಿ/ಕಡಲೆ	Kharif/Rabi	Fix nitrogen; good for rotation.

2. Crop Diseases & Pest Management

2.1 Tomato — yellowing leaves (very common question)

Yellow tomato leaves usually mean one of these — diagnose by pattern:

Pattern	Likely cause	Action
Older/lower leaves yellow first, even color	Nitrogen deficiency	Apply urea/compost or 19:19:19 foliar spray.
Yellow between veins, veins stay green	Magnesium / iron deficiency	Spray Epsom salt (MgSO_4) 1% or micronutrient mix.
Yellowing + wilting one side, brown stem inside	Fusarium / bacterial wilt	Remove plant; rotate crop; use resistant varieties; soil drench.
Curling + yellow + stunted, whiteflies present	Tomato leaf curl virus (spread by whitefly)	Control whitefly (yellow sticky traps, neem oil); remove infected plants.
Yellow with brown spots / concentric rings	Early blight (Alternaria)	Spray Mancozeb; remove affected leaves; avoid overhead watering.
Overwatered, soggy soil	Root suffocation	Improve drainage; reduce irrigation.

2.2 Other common diseases

- **Paddy blast:** diamond-shaped grey spots on leaves. Use resistant varieties, balanced N (avoid excess), spray Tricyclazole.
- **Paddy bacterial leaf blight:** leaf tips yellow then dry. Drain field, avoid excess nitrogen, use certified seed.
- **Cotton bollworm:** holes in bolls. Use pheromone traps, Bt cotton, need-based spraying.
- **Powdery mildew (many crops):** white powder on leaves. Spray wettable sulphur or neem.
- **Downy mildew:** yellow patches with greyish underside. Improve airflow, spray Metalaxyl + Mancozeb.
- **Aphids / whitefly / thrips:** sap-suckers, cause curling and virus spread. Neem oil 5 ml/L, yellow/blue sticky traps, Imidacloprid if severe.

Always start with cultural + organic controls (sanitation, traps, neem). Use chemical pesticides only as needed and follow label dosage and pre-harvest intervals.

3. Fertilizers & Nutrient Management

3.1 The three main nutrients (NPK)

Nutrient	Role	Deficiency sign	Common source
Nitrogen (N)	Leaf & vegetative growth	Old leaves yellow, stunted	Urea (46% N), compost

Phosphorus (P)	Roots, flowering	Purple/dark leaves, poor roots	DAP, SSP
Potassium (K)	Fruit quality, disease resistance	Leaf-edge scorching	MOP (potash)

3.2 Practical guidance

- Apply fertilizer based on a **Soil Health Card** recommendation — don't over-apply urea.
- Split nitrogen into 2–3 doses (basal + top-dressing) for better uptake.
- Use **organic manure** (FYM, vermicompost) to improve soil structure and water holding.
- **Biofertilizers**: Rhizobium (pulses), Azospirillum, PSB reduce chemical need.
- Micronutrients (Zn, Fe, B) matter for tomato, paddy, pulses — correct with foliar sprays.

4. Irrigation & Water Management

- **Drip irrigation**: saves 40–60% water, ideal for tomato, cotton, sugarcane, orchards. Subsidies available (see schemes).
- **Sprinkler**: good for groundnut, pulses, vegetables on undulating land.
- **Paddy**: alternate wetting & drying (AWD) saves water vs continuous flooding.
- Irrigate **early morning or evening** to reduce evaporation.
- Mulching (straw/plastic) conserves moisture and suppresses weeds.
- Watch for over-irrigation — it causes root rot, yellowing, and nutrient leaching.

5. Soil Health

- Get a **Soil Health Card** (free govt scheme) every 2–3 years for N-P-K and pH.
- Ideal pH for most crops is 6.0–7.5. Acidic soil → add lime; alkaline → add gypsum.
- Practice **crop rotation** (cereal → legume) to restore nitrogen and break pest cycles.
- Green manure (sunhemp, dhaincha) ploughed in boosts organic matter.
- Avoid burning crop residue — incorporate it or compost it.

6. Government Schemes (Central + Karnataka)

Scheme	Benefit	For whom
PM-KISAN	₹6,000/year in 3 installments of ₹2,000 (direct to bank)	Landholding farmer families
PM Fasal Bima Yojana (PMFBY)	Crop insurance against natural calamity; low premium (2% Kharif, 1.5% Rabi)	All farmers, esp. loanees
Kisan Credit Card (KCC)	Low-interest crop loans up to ₹3 lakh	All cultivators

Soil Health Card	Free soil testing & nutrient advice	All farmers
PM-KUSUM	Subsidy for solar pumps & solar power	Farmers with irrigation needs
Per Drop More Crop (PMKSY)	Subsidy on drip/sprinkler micro-irrigation	All farmers
Raitha Siri (Karnataka)	Incentive for millet (ragi/jowar) growers	Karnataka millet farmers
Krishi Bhagya (Karnataka)	Support for rainwater harvesting / farm ponds	Rainfed farmers in Karnataka
Raitha Samparka Kendra	Local hub for seeds, advice, scheme enrolment	All Karnataka farmers

Scheme amounts/rules change over time. Advise farmers to confirm at the official portal (pmkisan.gov.in, etc.) or the nearest Raitha Samparka Kendra / agriculture office.

7. Market Information & MSP

- **MSP (Minimum Support Price):** government-guaranteed price for notified crops (paddy, ragi, jowar, tur, cotton, groundnut, etc.), announced each season by CACP.
- **eNAM (National Agriculture Market):** online platform to sell produce across mandis for better prices — enam.gov.in.
- **APMC mandis:** regulated markets; check daily prices before selling.
- **FPOs (Farmer Producer Organizations):** group selling gets better rates and bargaining power.
- **Tips:** grade and clean produce, sell in lots, track price trends, avoid distress sale right after harvest if storage is possible.

8. Seasons & Calendar

- **Kharif (monsoon):** sow Jun–Jul, harvest Sep–Oct. Paddy, ragi, maize, cotton, tur, groundnut.
- **Rabi (winter):** sow Oct–Nov, harvest Feb–Mar. Wheat, gram, jowar, sunflower.
- **Summer/Zaid:** short irrigated crops — vegetables, groundnut, fodder.

9. Quick Q&A (sample interactions)

Q: My tomato plant leaves are turning yellow. What should I do?

A: First check the pattern. If older leaves yellow evenly, it's likely nitrogen shortage — add compost or urea. If yellowing is between the veins, spray Epsom salt (magnesium). If leaves

also curl and you see tiny white insects, it's leaf-curl virus from whiteflies — set yellow sticky traps, spray neem oil, and remove badly affected plants. Avoid overwatering and don't wet the leaves in the evening.

Q: How much urea should I apply to paddy?

A: Follow your Soil Health Card. As a general guide, apply nitrogen in 3 splits — at transplanting, tillering, and panicle initiation — rather than all at once. Don't overdo urea: excess nitrogen invites blast and pest attack. Confirm exact quantity with your local Raitha Samparka Kendra.

Q: Which scheme gives money directly to farmers?

A: PM-KISAN gives ₹6,000 per year in three ₹2,000 installments straight to your bank account. You need to register with your land records and Aadhaar at pmkisan.gov.in or your nearest agriculture office.

Q: How can I save water for my crop?

A: Use drip irrigation (there's a government subsidy for it), water early morning or evening, and mulch the soil with straw to hold moisture. For paddy, try alternate wetting and drying instead of keeping the field always flooded.

Disclaimer: This knowledge base is general guidance for an AI assistant demo. Pesticide dosages, scheme amounts, and MSP values change — farmers should verify with official sources (KVK, Raitha Samparka Kendra, government portals) before acting.